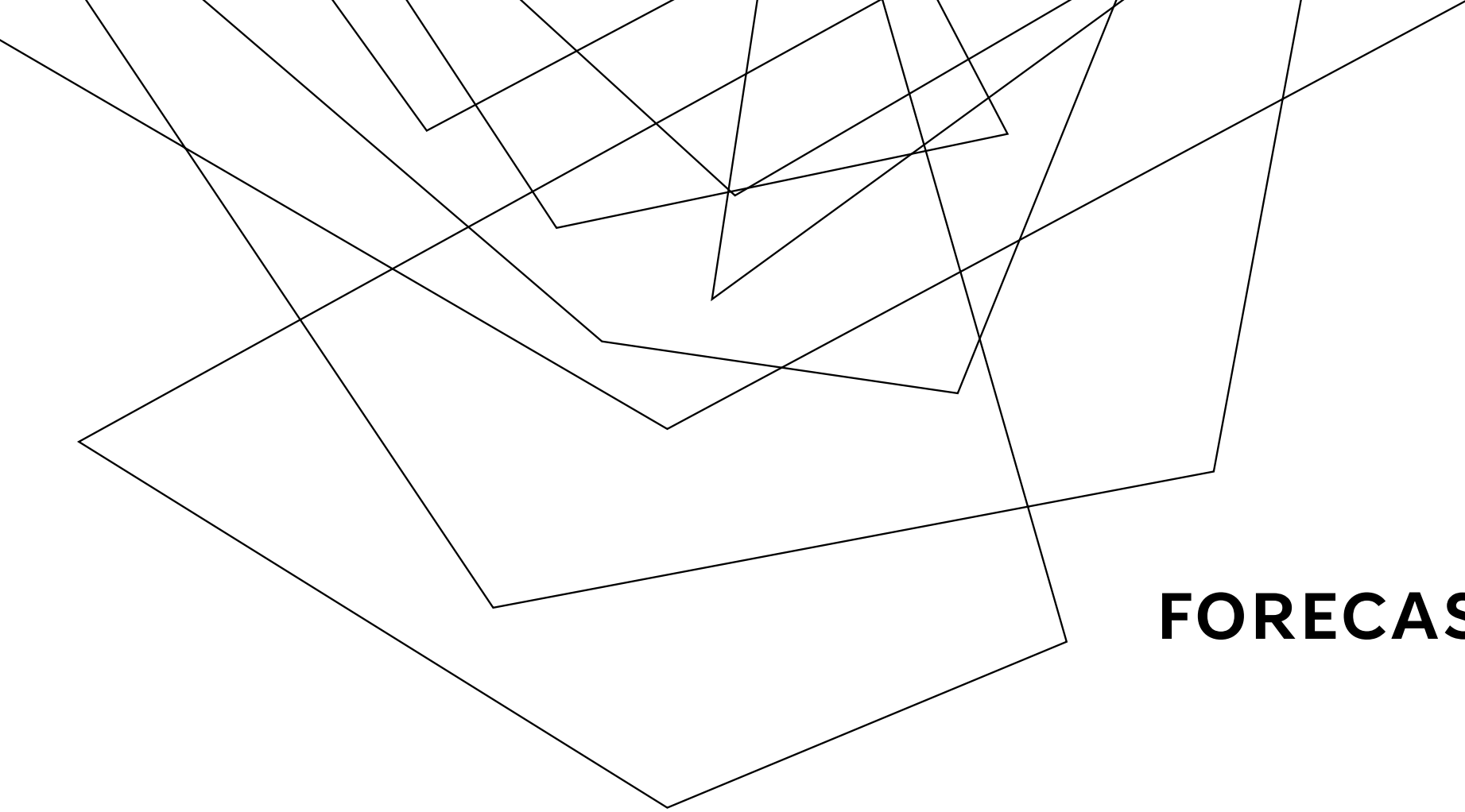


A series of thin, black, overlapping geometric lines and polygons in the top-left corner of the slide, creating a complex, abstract pattern.

# FORECASTING LMP

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# OUTLINE

1. Introduction: Why It Matters?
2. Problem Statement
3. Exploratory Data Analysis (EDA)
4. Model & Methodology
5. Validation & Experiments
6. Conclusion & Future Work
7. Lessons learnt

# INTRODUCTION

## Forecasting Day-Ahead Electricity Prices: Why It Matters?

### Background & Motivation

- Challenge in the **deregulated** energy markets. (Why?) → Renewable energy integration and fuel price fluctuations make electricity prices highly unstable.
- Transition to deregulated markets enhanced the grid reliability by encouraging stakeholders to optimize the bidding strategies and manage operational output.
- Limitation of deregulated markets: → Introduced unique electricity price variability
- This limitation raised the notion of Location Marginal Price (LMP) forecasting
- LMP quantifies the marginal cost to deliver an additional unit of electricity at a specific location (or node is the grid network)
- Accurate LMP forecasts can cut imbalance penalties by up to **20%**, thus saving market participants millions annually in hedging and operational costs [1].

Accurate LMP  $\hat{y}_t$  = Efficient Operations + Lower Utilities Cost & Energy Generation

# INTRODUCTION

## Forecasting Day-Ahead Electricity Prices: Why It Matters?

LMP forecasts regression modeling formalism (Autoregressive models):

$$y_{LMP,t} \sim \text{SARIMAX with } f \begin{pmatrix} y_{Load_{t-1}}, y_{Solar_{t-1}}, y_{Wind_{t-1}}, y_{FuelPrice_{t-1}}, y_{AirTemp_{t-1}} \\ \vdots \\ y_{Load_{t-p}}, y_{Solar_{t-p}}, y_{Wind_{t-p}}, y_{FuelPrice_{t-p}}, y_{AirTemp_{t-p}} \end{pmatrix}$$

What do you think is the weakest regressor? Still include it? Why?

Despite tuning SARIMA-X, two challenges persist:

1. Models shows inherent heteroscedasticity of LMP
2. Non-Gaussian distribution of the forecast residuals

# PROBLEM STATEMENT

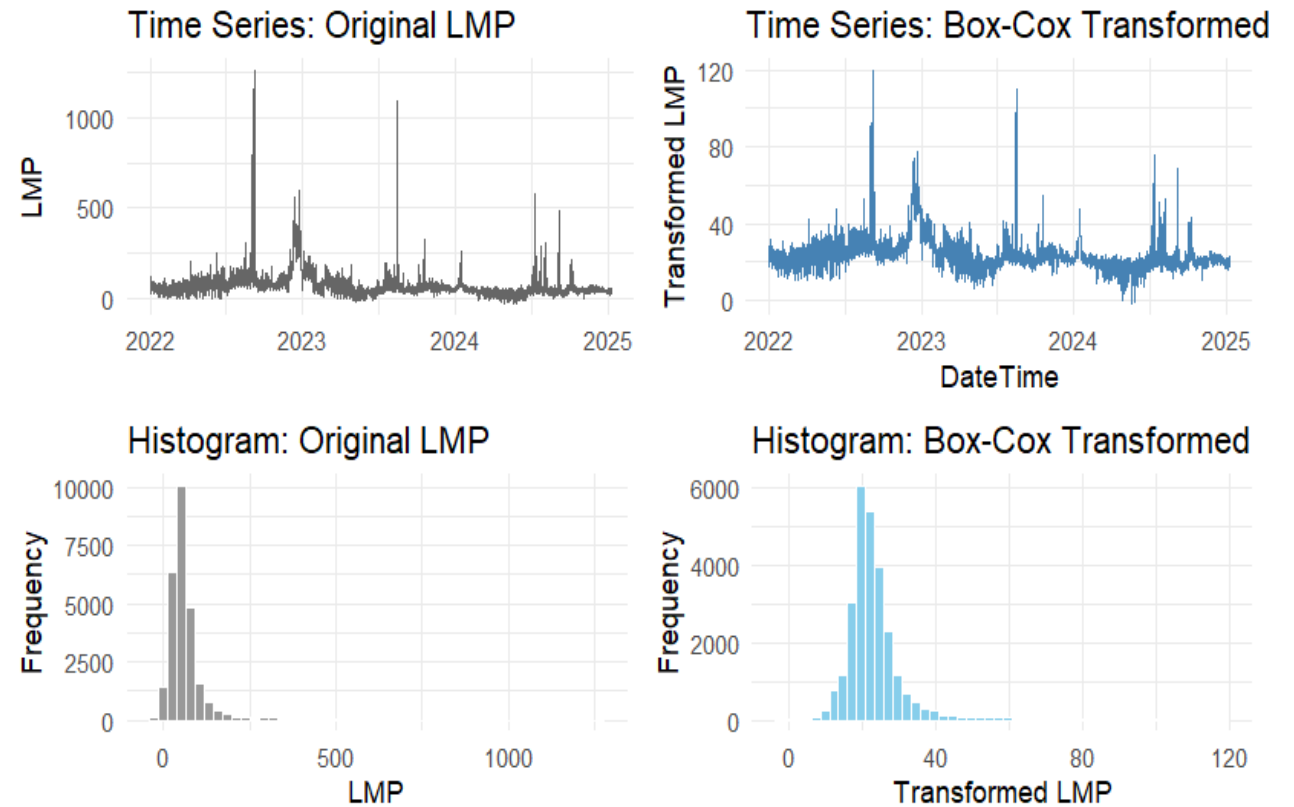
- Variance stabilization.

A variance stabilizing transformation (in our case, Box-Cox) is used to align the residual behavior with Gaussian error structures and improve parameter estimation.

- Model parsimony vs. reliability.

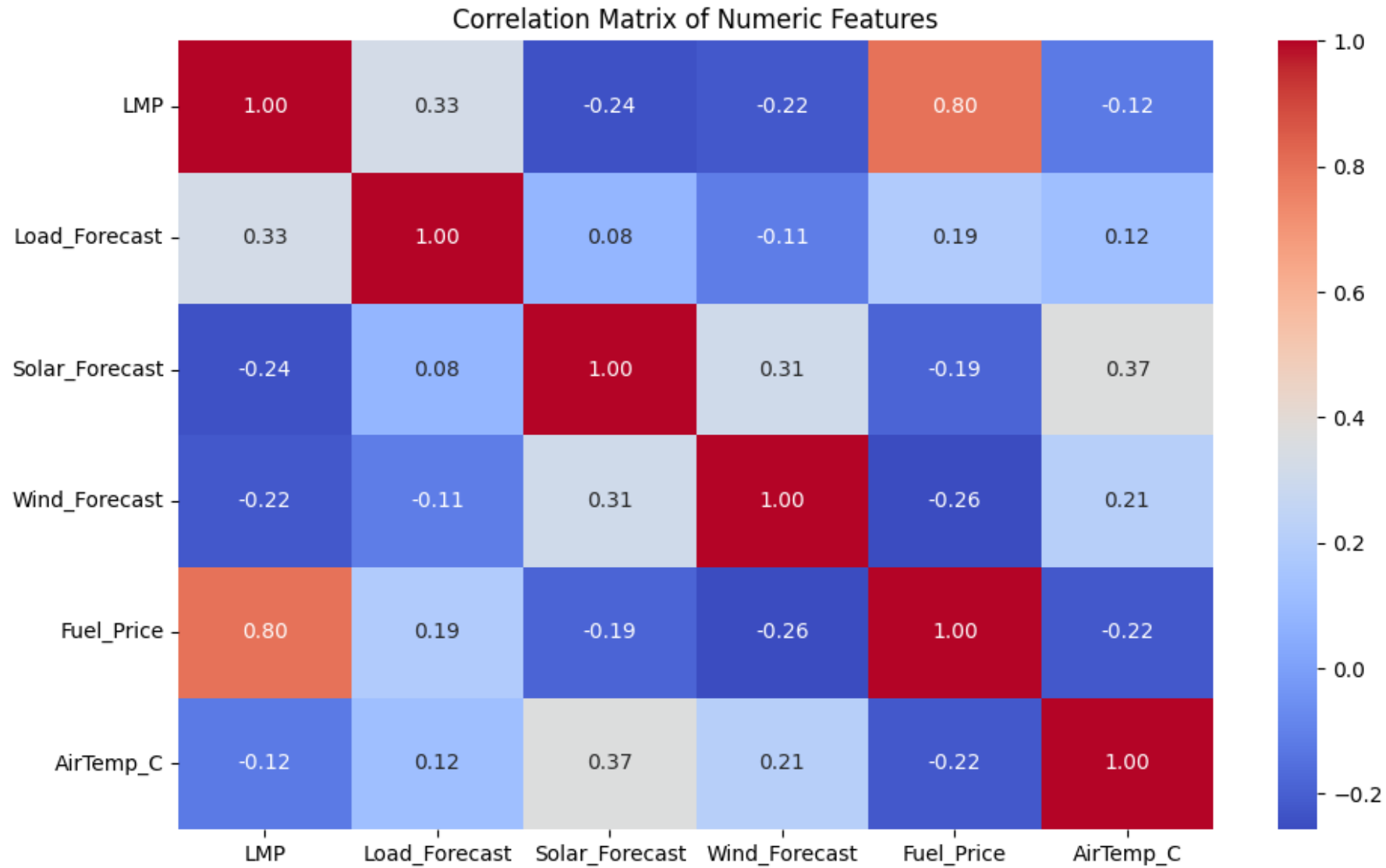
Engineered covariates such as dummy variables, lagged regressors, or seasonal differencing can improve the model fit but risk overfitting and increased run-time.

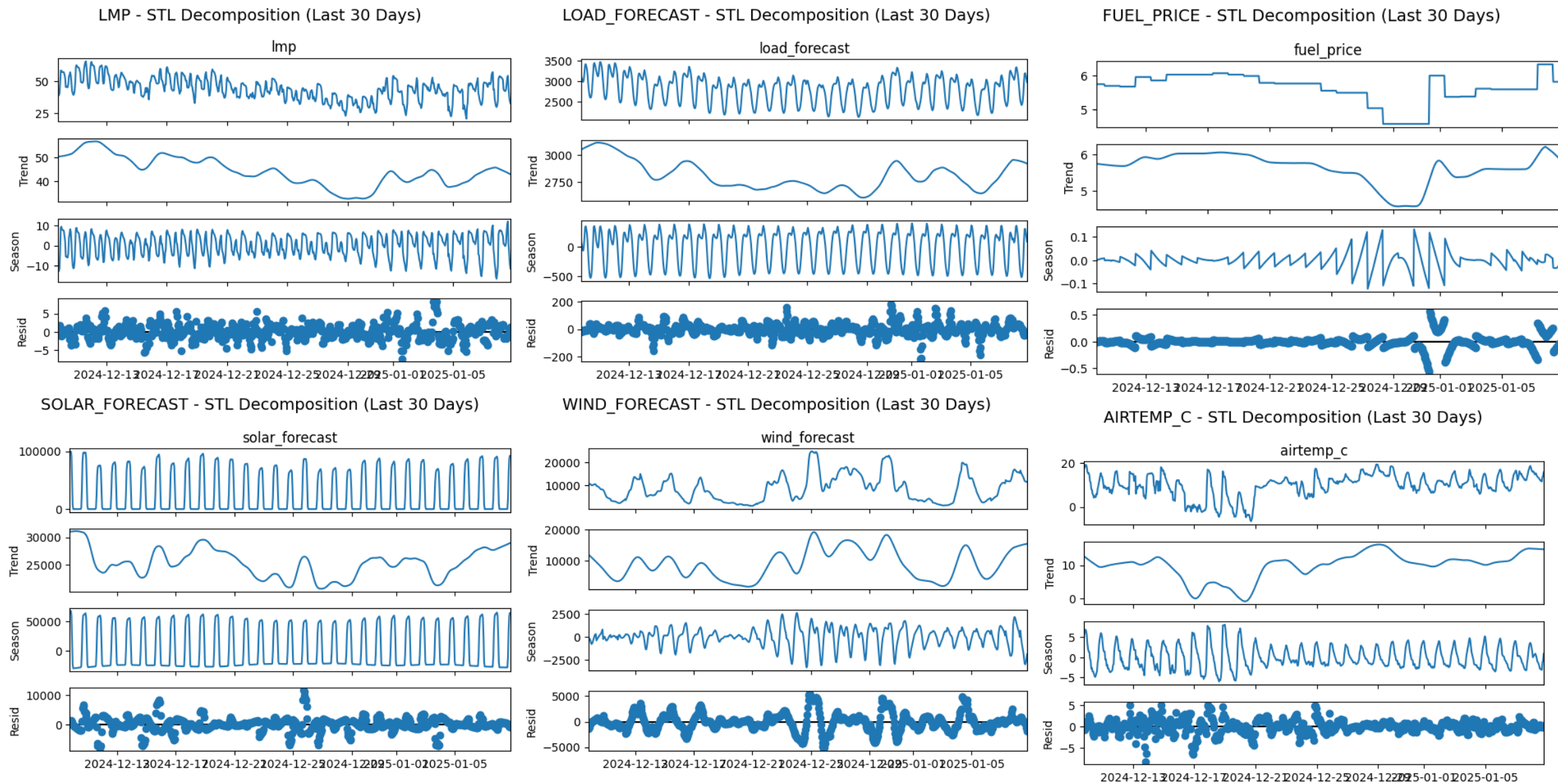
Therefore, adding Box-Cox  $\lambda$  estimation must preserve the model parsimony, while strengthening the forecast stability.



*Figure 1 Time Series Plots & Histograms of Original LMP vs. Box-Cox Transformed LMP*

# EDA





**EDA** Last 30 days (from 2024-12-09 19:00:00 to 2025-01-08 19:00:00)

# EDA

Full time series of Solar and Wind:

- Sudden, significant increase, roughly ten-fold jump in magnitude, beginning in mid-April/early May 2024.
- Change in the upstream data scaling rather than a genuine surge in renewable output.
- Rescale all post-April 13, 2024, values by dividing them by 10 within both the training and testing sets.

Restore continuity in the covariates scales to preserve the integrity of the  $\lambda$  estimation.

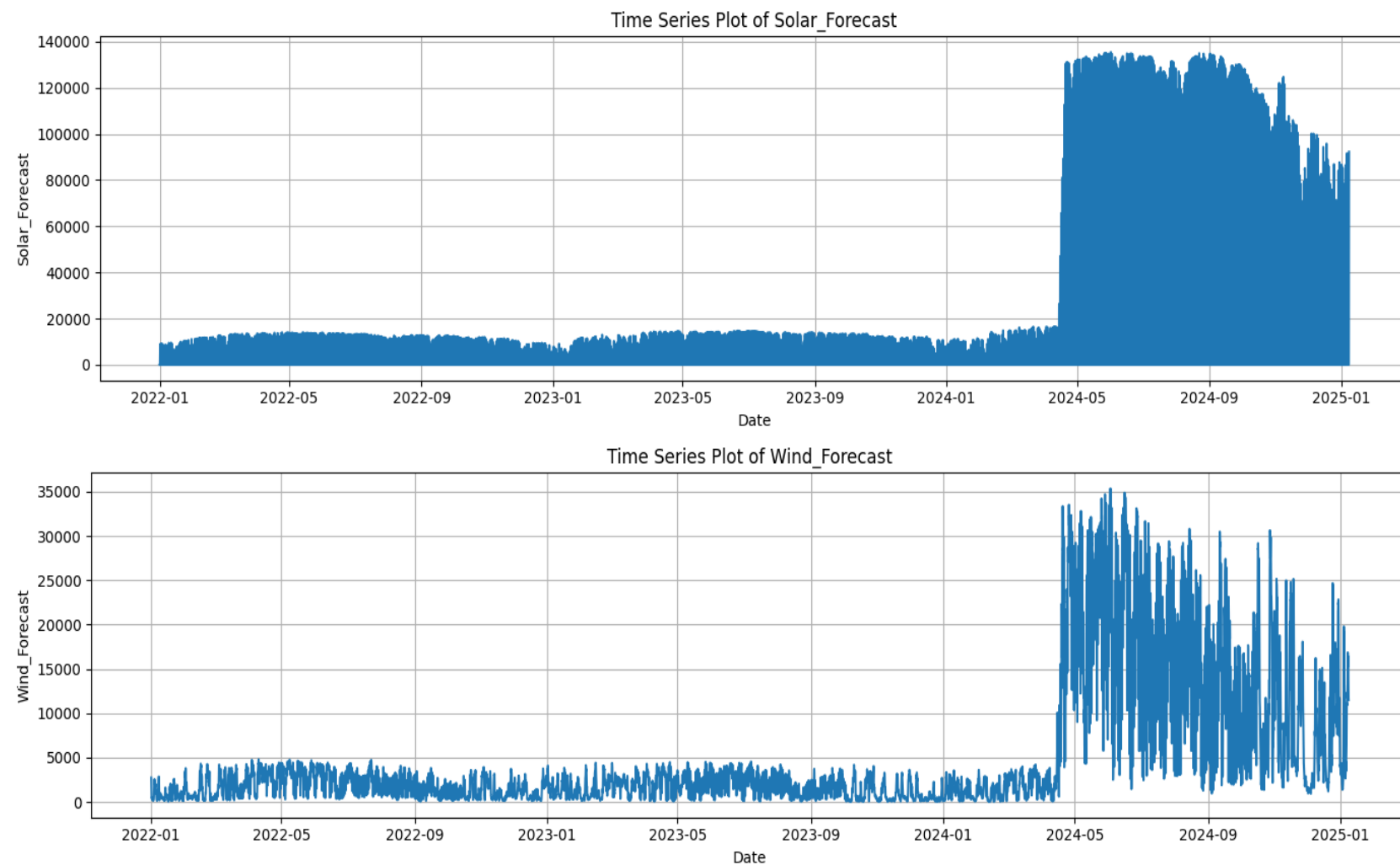


Figure 3 Time Series Plots (full training data) of: (a) Top plot: Solar Forecast Time Series and (b) Bottom plot: Wind Forecast Time Series



# MODEL & METHODOLOGY

- Data analysis motivated the implementation of a model that can handle exogenous features
- Statistical Models:
  - SARIMA-X, DHR
  - Used PACF analysis to pick the autoregressive parameter
  - Trial and error with residual analysis for moving average terms
  - Seasonal differencing for stationarity
- ML models:
  - XGB, LGBM
  - Conducted feature engineering to handle temporal features and better recognize seasonality

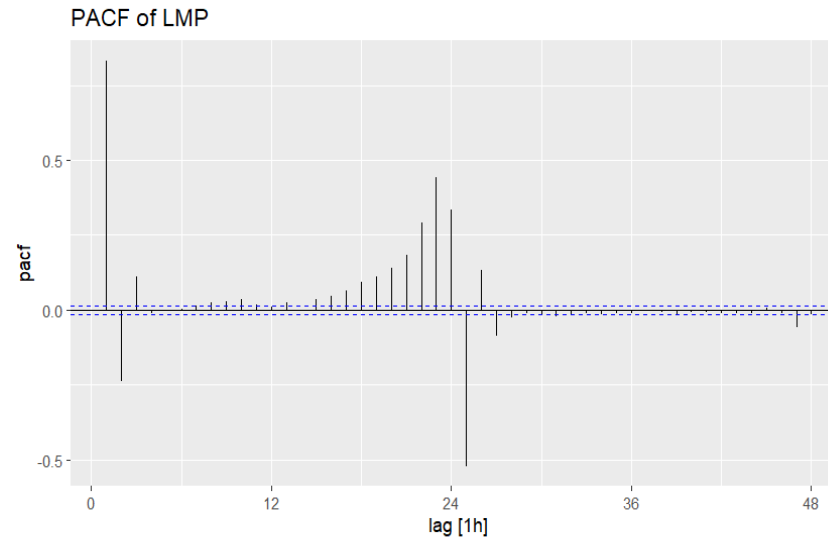


Figure 3 The PACF plot for the LMP training data

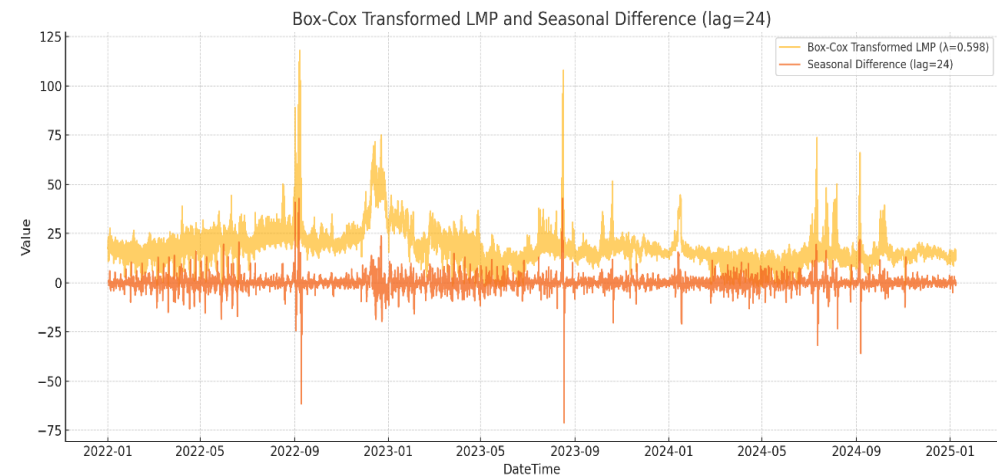
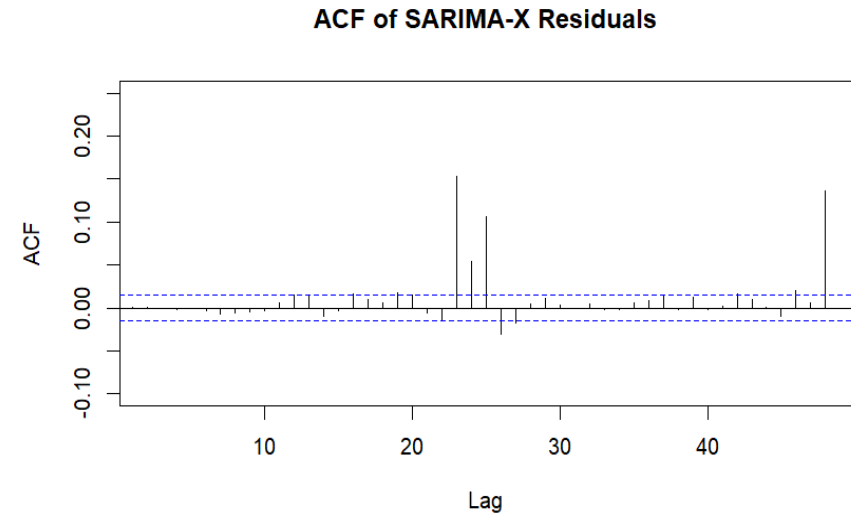


Figure 4 Superimposed Time Series Plot of LMP (Box-Cox  $\lambda = 0.598$  Transformed) vs. LMP (Seasonal Differenced) with 24-lag

# VALIDATION & EXPERIMENTS

- Benchmark Models
- Residual Analysis
  - Analyzed residual timeseries plot, ACF plot, and histogram
  - Looked for white noise signature
- Rolling Testing
  - 24-hour rolling test window for 37 days
  - Did not retrain the model each day, but updated the input data (time/computational constraints)



*Figure 5 The ACF plot of the residuals from the chosen and fitted SARIMA-X model*

| Forecast Model | MAE  | % MAE Improvement on Naïve |
|----------------|------|----------------------------|
| Naïve          | 5.79 | N/a                        |
| Seasonal Naïve | 4.28 | 26.08%                     |
| XGB            | 3.98 | 31.26%                     |
| LGBM           | 4.02 | 30.57%                     |
| SARIMA-X       | 3.09 | 46.65%                     |

# CONCLUSION & FUTURE WORK

- SARIMA-X regression formalism to fit a model that is complex enough to correctly represent and forecast LMP values with a good accuracy
- Integrating a Box-Cos transformation ( $\lambda \approx 0.598$  using Guerrero's method) to stabilize the variance and normalize the residuals.

Single SARIMA(3,0,2)(0,1,2)[24] model, augmented by five regressors and transformed for variance stability, can capture both the cyclical and unstable dynamics in day-ahead LMP forecasting.

Future work:

- Evaluate a fully probabilistic forecasts by implementing and comparing interval scoring metrics, e.g., QS, WS, or CRPS, to assess both calibration and sharpness of the LMP prediction intervals.
- Develop regime-specific Box-Cox transformation by estimating  $\lambda$  using maximum likelihood over a detected price regimes → Stabilize variance around the structural breaks.
- VARIMA-X multivariate model (jointly modeling several interconnected LMP nodes with exogenous predictors, specifically load and renewables, to develop spatial and cross-series correlations.

# LESSONS LEARNT

- Tradeoff: Model Complexity vs. Training Time
- Scientific Process is Crucial
- Risks of Forecasting on Forecasts
- Application-based Learning Reinforced Theory
- Takeaways:
  - Adaptability and persistence are key
  - Preconceived thoughts must be validated using data



# REFERENCES

[1] H. Zareipour, C. A. Canizares, and K. Bhattacharya, “Economic impact of electricity market price forecasting errors: A demand-side analysis,” *IEEE Transactions on Power Systems*, vol. 25, no. 1, pp. 254–262, Mar. 2010, doi: 10.1109/TPWRS.2009.2030380



**THANK YOU!**